Imperial College London

MRes EXAMINATION June 2024

STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students
June 5th, 2024: 10:00 to 12:00

Answer all FOUR questions
Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. A set of data points $\{x_i, y_i\}_{i=1}^{N}$ are recorded in an experiment, where x_i are the independent variables and y_i are the dependent variables. A colleague suggests that the relationships between the data should be well described by a second order polynomial of the form,

$$y(x) = a_0 + a_1 x + a_2 x^2.$$

You are told that the measured values x_i are known perfectly, but the y_i values can each be described as an independent Gaussian random variable with known standard deviation $\delta > 0$,

$$y_i \sim \frac{1}{\delta\sqrt{2\pi}} \exp\left(-\frac{[y_i - y(x_i)]^2}{2\delta^2}\right).$$

for $i = 1, 2, \ldots, N$.

(i) Write down the likelihood function $L(a_0, a_1, a_2)$ for the data points $\{x_i, y_i\}_{i=1}^{N}$.
[1 mark]

(ii) Derive a matrix equation that could be solved to find the maximum likelihood estimators $\hat{a}_0, \hat{a}_1, \hat{a}_2$ of the coefficients a_0, a_1, a_2 in terms of the data points $\{x_i, y_i\}_{i=1}^{N}$. You do not need to solve this equation.

HINT: Start by writing down the negative log-likelihood function: $-\ln L(a_0, a_1, a_2)$.
[9 marks]

(iii) Take a look at figure 1 which shows a set of 10 data points $\{x_i, y_i\}_{i=1}^{N}$ and the best fit second order polynomial $y(x) = \hat{a}_0 + \hat{a}_1 x + \hat{a}_2 x^2$ to the data (solid line) obtained by finding the maximum likelihood estimators. The error bars on each point show the known standard deviation (δ). A second colleague suggests that the relationship between the y and x values could be better described by a simple linear relationship of the form,

$$y(x) = b_0 + b_1 x.$$

They also calculated the maximum likelihood estimators $\hat{b}_0$, $\hat{b}_1$ for this model and the resulting linear relationship is shown in figure 1 as a dashed line.

(a) Find the maximum likelihood estimators for the parameters b_0 and b_1.
HINT: the linear relationship is equivalent to the polynomial relationship when $a_2 = 0$.
[6 marks]

(b) Devise a hypothesis test to determine if the best linear model, defined by the maximum likelihood estimates $\hat{b}_0$, $\hat{b}_1$ shown by the dashed line in figure 1, should be rejected in favour of the best second order polynomial model defined by the maximum likelihood estimates $\hat{a}_0$, $\hat{a}_1$, $\hat{a}_2$ and shown by the solid line in figure 1. You must clearly state the null and alternative hypotheses, the test statistic that you would use, the critical region and size of the test and a full procedure for determining the outcome of the test.
[7 marks]

(c) Define the power $(1 - \beta)$ of a hypothesis test and explain how you would be able to calculate it in your answer to the previous question.
[2 marks]

[Total 25 marks]

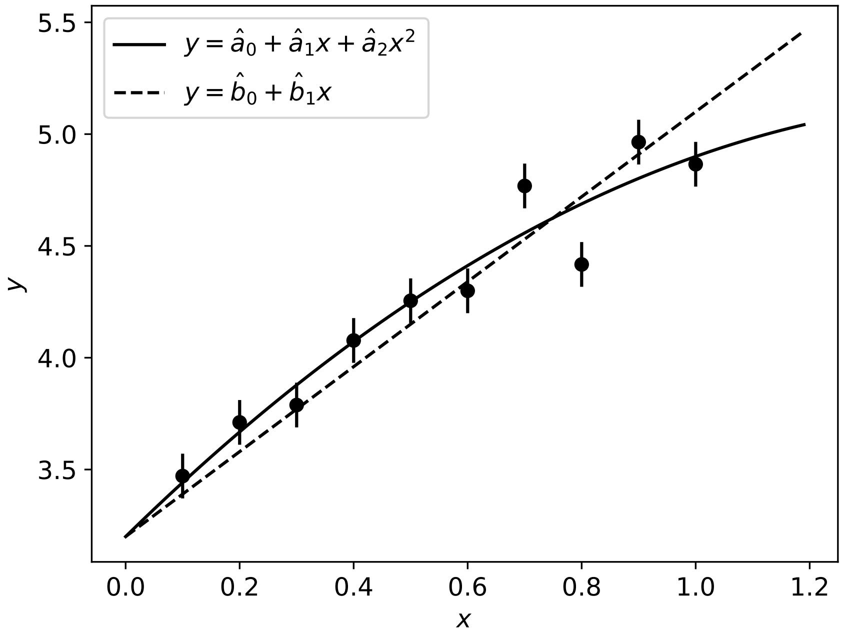
$y = \hat{a}_0 + \hat{a}_1 x + \hat{a}_2 x^2$
$y = \hat{b}_0 + \hat{b}_1 x$
5.5
5.0
4.5
4.0
3.5
y
0.0
0.2
0.4
0.6
0.8
1.0
1.2
x

Figure 1

2. Imagine a telescope measuring the positions of stars in a small region of the sky. The telescope has an unknown position resolution σ that needs to be determined before being put into operation. The telescope is aligned so that the origin $(0, 0)$ is centered on the apparent position of the star (shown in figure 1), while the true position is unknown. The X and Y co-ordinates of the difference between the true and apparent positions will be random normal distributed variables with mean 0 and standard deviation σ.

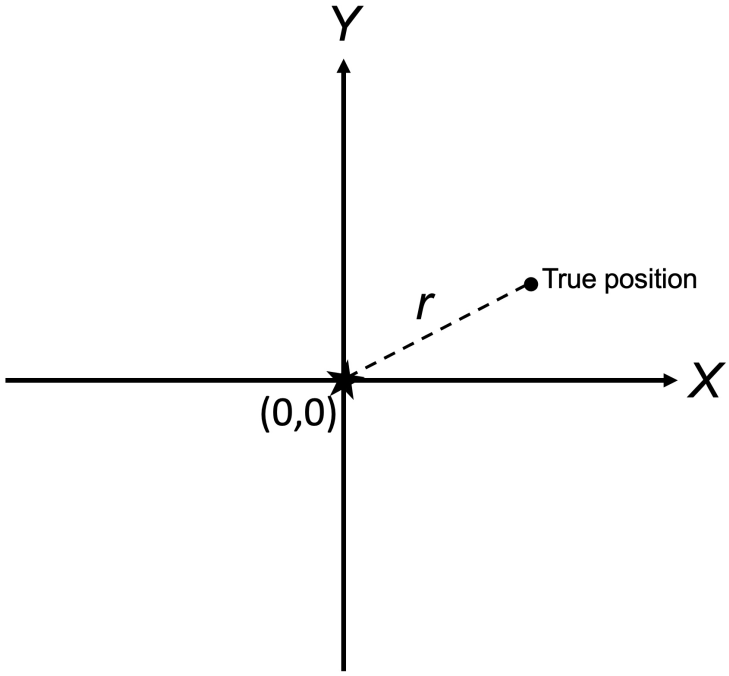

Figure 1

The random variable $r = \sqrt{X^2 + Y^2}$ will be distributed as a Rayleigh distribution, with a probability density given by,

$$f(r; \sigma) = A(\sigma) r e^{-\frac{r^2}{2\sigma^2}},$$

where A is a normalisation term depending on σ. The variable will always be a positive real value $r \in [0, +\infty)$.

(i) Show that the normalisation term $A(\sigma)$ is given by,

$$A(\sigma) = \frac{1}{\sigma^2}.$$

[2 marks]

(ii) Calculate the second algebraic moment $\mu_2 = E\left[r^2\right]$ of the Rayleigh distribution. You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

[3 marks]

[This question continues on the next page . . .]

(iii) A set of calibration data $\{r_i\}_{i=1}^{N}$ using known positions of N stars is used to estimate the resolution of the telescope. These measurements should be assumed to be independent and identically distributed.

(a) Using the method of moments, find the expression for an estimator of σ^2 using the calibration data from the second algebraic moment. [2 marks]

(b) Is the estimator you determined a biased or un-biased estimator? [5 marks]

(c) Write down the likelihood function $L(\sigma)$ for the observations $\{r_i\}_{i=1}^{N}$. [1 mark]

(d) Show that the maximum likelihood estimator $\hat{\sigma}$ for σ is given by,

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{N} r_i^2}{2N}}.$$

HINT: Start by finding $-\ln L(\sigma)$. [5 marks]

(e) Show that the maximum likelihood estimator $\hat{\sigma}$ is a *biased* estimator of σ. You may use the fact that for $g(x) = \sqrt{x}$, the expectation satisfies $E[g(x)] < g(E[x])$. [4 marks]

(iv) Aside from being an unbiased estimator, name one other property of a *good* estimator and describe what that property means. You can use equations and/or diagrams in your answer but the explanation must be clearly stated. [3 marks]

[Total 25 marks]

3. (i) (a) A simple drug trial involves giving N patients the drug and recording how many respond positively to treatment. Assuming that the outcomes of each patient are independent, and the probability of a successful outcome is λ, argue that the sampling distribution for n, the number of patients who respond positively to treatment, is the binomial distribution

$$p(n|N,\lambda) = \frac{N!}{n!(N-n)!}\lambda^n(1-\lambda)^{N-n}.$$

[4 marks]

(b) Write down an unnormalised posterior distribution for λ, given n successful outcomes from a fixed number of patients N. Justify your choice of prior.

[3 marks]

(ii) (a) Find the maximum posterior value for λ for given n and N. [3 marks]

(b) Show that the posterior mean of λ for general n and N is

$$\bar{\lambda} = \frac{n+1}{N+2}.$$

You may assume the following without proof:

$$\int_0^1 x^a(1-x)^b dx = \frac{\Gamma(1+a)\Gamma(1+b)}{\Gamma(2+a+b)},$$

and $\Gamma(z+1) = z\Gamma(z)$.

[7 marks]

(iii) (a) What is meant by the term *conjugate prior*? [2 marks]

(b) Show that the Beta distribution, $\pi(\lambda) \propto \lambda^x(1-\lambda)^y$ is a conjugate prior for the binomial distribution, and find an expression for the posterior parameters in terms of x, y, n and N.

[6 marks]

[Total 25 marks]

4. (i) Write notes on how simulation-based inference (SBI) can be used to estimate Bayesian posteriors, using a simple example with one parameter and one data point for illustration. Using sketches, also show how the sampling distribution may be estimated. Discuss in particular Approximate Bayesian Computation (ABC), and the challenges of SBI. [10 marks]

(ii) An experiment consists of 10 independent trials, each of which is either successful or unsuccessful. The number of successes is 6, and the probability of success is a parameter θ that is to be inferred. A simulator draws 100 θ values from a uniform prior between 0 and 1. The parameters and number of successes in a short simulation run is given in the table below. Estimate the posterior, in the form of a histogram in five bins with boundaries at 0, 0.2, 0.4, 0.6, 0.8 and 1.0, using ABC. [5 marks]

θ	n	θ	n	θ	n	θ	n
0.25	3	0.22	3	0.18	2	0.54	6
0.97	9	0.05	0	0.34	5	0.37	4
0.80	7	0.81	6	0.46	3	0.75	7
0.61	6	0.39	3	0.66	5	0.07	1
0.97	9	0.87	9	0.78	8	0.82	9
0.81	9	0.49	4	0.77	9	0.57	8
0.25	2	0.71	5	0.76	5	0.18	0
0.61	7	0.99	10	0.33	3	0.77	7
0.85	9	0.69	8	0.69	5	0.83	8
0.91	10	0.40	4	0.26	0	0.14	1
0.74	3	0.59	6	0.37	6	0.45	2
0.31	2	0.40	4	0.90	7	0.28	4
0.93	10	0.42	5	0.39	3	0.01	1
0.75	8	0.51	6	0.84	7	0.34	5
0.64	6	0.73	7	0.75	7	0.95	10
0.45	4	0.65	8	0.73	9	0.71	7
0.94	9	0.80	7	0.22	3	0.87	10
0.30	4	0.05	1	0.07	0	0.11	0
0.55	6	0.73	8	0.71	6	0.89	7
0.94	9	0.38	3	0.20	1	0.94	10
0.32	6	0.71	8	0.53	6	0.07	0
0.44	2	0.62	6	0.05	1	0.65	7
0.36	4	0.09	1	0.41	4	0.78	7
0.47	4	0.16	3	0.37	6	0.29	2
0.46	6	0.56	6	0.65	5	0.63	4

(iii) An important part of SBI is the need for extreme data compression. An experiment records measurements $x_i; i = 1, \ldots N$, drawn independently from a distribution with mean μ, and known gaussian errors σ_i.

(a) Write down the likelihood, and show that it can be factorised into three terms, which are (a) a function of x_i only; (b) a function of μ only; (c) a term that includes both x_i and μ. (σ_i^2 will appear in all of them). [4 marks]

(b) Argue that for the posterior of μ, we can ignore the term that depends on x_i only. [3 marks]

(c) By considering the other terms, show that we can compress the x_i into a single number

$$X \equiv \sum_{i=1}^{N} \frac{x_i}{\sigma_i^2}$$

and still compute the posterior exactly. [3 marks]

[Total 25 marks]